

Algebra II with Trigonometry

Chapter 7.1

Mental Math (Gemini)

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Quotient Property 1. What does the fraction $\sin(x) / \cos(x)$ simplify to?

Product Property 2. According to the magic hexagon's product properties, what is the result of multiplying $\tan(x)$ by $\cos(x)$?

Pythagorean Identity 3. What is the exact numerical value of the expression $\sin^2(x) + \cos^2(x)$?

Reciprocal Property 4. Which trigonometric function is equivalent to $1 / \csc(x)$?

Cofunction Identity 5. What does the cofunction expression $\tan(\frac{\pi}{2} - x)$ simplify to?

Rearranged Pythagorean Identity 6. If you subtract 1 from $\sec^2(x)$, what trigonometric expression is left?

Even-Odd Identity 7. Based on the even-odd identities used in the examples, does $\cos(-x)$ simplify to $\cos(x)$ or $-\cos(x)$?

Basic Simplification 8. By rewriting the expression in terms of sine and cosine, what does $\sin(\theta) \sec(\theta)$ simplify to?

Diagonal Reciprocals 9. What is the product of $\tan(x)$ and $\cot(x)$?

Trigonometric Substitution Concept 10. If you substitute $x = \sec(\theta)$ into the expression $\sqrt{x^2 - 1}$, what single trigonometric function does the radical simplify to? (Assume $0 < \theta < \frac{\pi}{2}$)